Should Ethan Hunt Be Dead?

A survival model of the Mission: Impossible stunts | Technical note, October 2026 | Draft: inputs not yet checked against footage

Abstract

We turn 34 stunt entries from seven Mission: Impossible films into per-event fatality probabilities using simple physical models for falls, wall impacts, vehicle strikes, explosions, breath-holds and fights, then multiply the survival chances across 154 individual hits. The point estimate for surviving every stunt is 1 in 1,716 (1 in 1,305 with no accumulated damage). A 5,000-trial uncertainty run gives a median of 1 in 64,393, with a 5th to 95th percentile range from 1 in 1,038 to 1 in 5.5 x 10^7. Vehicle collisions carry 77% of the total hazard, so the result mostly reflects my guesses about effective crash speeds. Even cutting every vehicle speed by a quarter leaves about 1 in 43. The model supports the claim that Hunt's career is implausible for an ordinary body, but it does not support any particular number.

1. Question and approach

Action-film heroes survive events that would kill most people. This note asks how unlikely the survival of one character is when each stunt is scored by a physical model instead of by intuition. Each stunt becomes a count of events and a per-event chance of death. Chances of survival multiply across events, and each film is treated as a block in which earlier injuries make later ones worse. The goal is a transparent order-of-magnitude estimate that others can change, not a forecast.

2. Data

The dataset lists 34 stunt entries from Mission: Impossible (1996), Mission: Impossible 2 (2000), Mission: Impossible III (2006), Ghost Protocol (2011), Rogue Nation (2015), Fallout (2018) and Dead Reckoning (2023). Each row gives the film, a short stunt description, the model that applies, two model inputs, an event count, a confidence level and a basis note. Inputs are my estimates from memory of the films, such as effective collision speed, charge size and distance, or hold time. Every row is marked unverified. Mission: Impossible: The Final Reckoning is excluded because I could not pin its stunt details down reliably.

3. Models of a single event

Each model returns p, the chance that one occurrence of the event is fatal. The parameters are stylized fits to widely quoted figures. They are not fitted to fresh data.

Falls and wall slams. Peak deceleration follows from height h (or impact speed v) and the stopping distance d of the surface:

$$g = h / d \text{ (fall)} \qquad g = v^2 / (2\, d\, g_0) \text{ (impact)}$$

$$p = 1 / (1 + \exp(-(\ln g - \ln 140) / 0.35))$$

The midpoint of 140 g corresponds to a fall of about 14 m onto a hard surface with 0.1 m of body give, which is near the commonly quoted four-storey fatality midpoint. Softer or more compliant landings raise d and lower g.

Vehicle strikes. A logistic curve in impact speed v in mph, shaped like published pedestrian-crash data:

$$p = 1 / (1 + \exp(-(v - 41.5) / 8))$$

Explosions. Peak overpressure comes from the Kinney-Graham scaling for a TNT-equivalent charge W (kg) at distance R (m), with $Z = R / W^{1/3}$:

$$P_s / P_0 = 808\, [1 + (Z/4.5)^2]\, /\, \mathrm{sqrt}([1 + (Z/0.048)^2]\, [1 + (Z/0.32)^2]\, [1 + (Z/1.35)^2])$$

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p = 1 / (1 + exp(-(ln P_s - ln 240 kPa) / 0.3))
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Breath-hold. Chance of a fatal blackout after t seconds, with half of blackouts assumed fatal:

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p = 0.5 / (1 + exp(-(t - 120) / 25))
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Fights and fixed-rate events. A flat 0.12% chance of a lethal head injury per blow. Events such as the lightning strike (10%), the BASE jump (about 1 in 2,300) and grip loss use a stated probability per event.

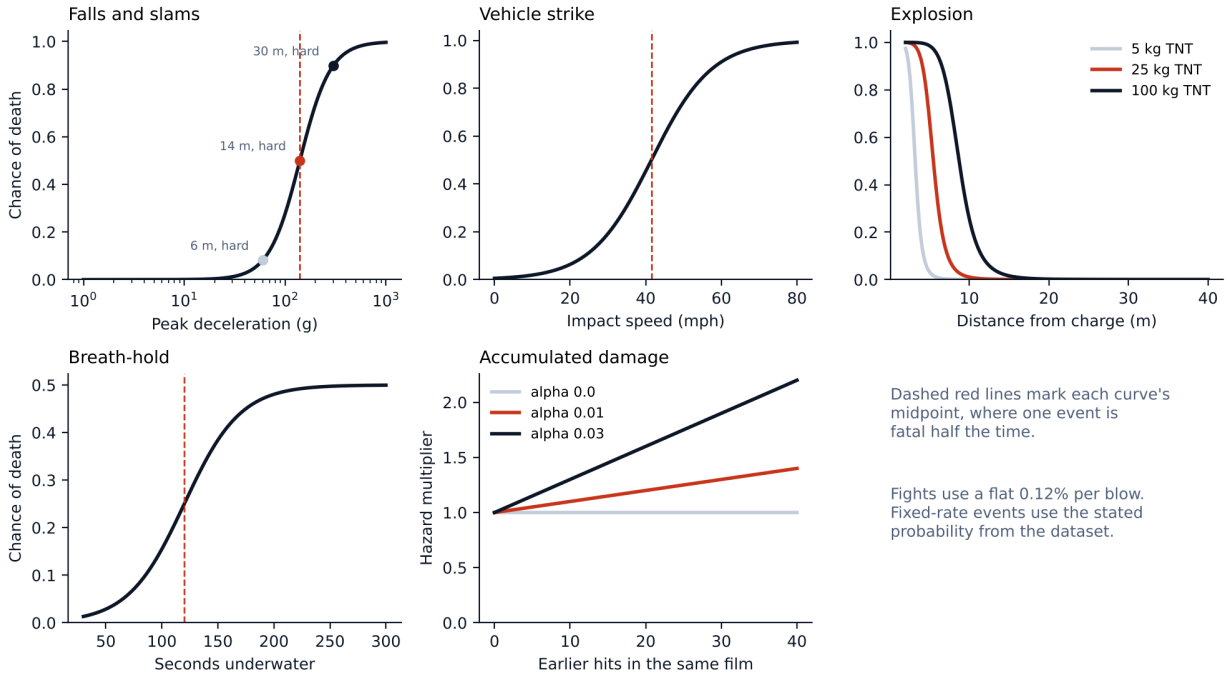


Figure 1. Single-event models. Dashed red lines mark each curve's midpoint. The bottom-middle panel shows the accumulated-damage multiplier.

4. Combining events

An event with fatality chance p has hazard h = -ln(1 - p). A film's events are processed in order. If k hits have already happened in the film, an event that occurs n times adds hazard

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H = h [ n (1 + a k) + a n (n - 1) / 2 ]  (a = accumulation rate)
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where α is the accumulated-damage rate (0.01 by default). Survival for the series is S = exp(-sum of H). With α = 0 the events are independent and the hazards simply add. For uncertainty, each trial multiplies speeds, distances, charge sizes and hold times by lognormal noise (sigma 0.25), multiplies fixed probabilities by lognormal noise (sigma 0.7), and jitters event counts (sigma 0.3). The run uses 5,000 trials and a seeded generator, so the JavaScript and Python implementations return identical numbers.

5. Results

Film	Survival	Odds against	Share of hazard
Mission: Impossible	95.9%	1 in 1	1%
Mission: Impossible 2	31.3%	1 in 3	16%
Mission: Impossible III	47.5%	1 in 2	10%
Ghost Protocol	37.0%	1 in 3	13%
Rogue Nation	10.3%	1 in 10	30%
Fallout	26.7%	1 in 4	18%
Dead Reckoning	40.1%	1 in 2	12%
All seven films	0.058%	1 in 1,716	100%

Table 1. Per-film and combined survival at alpha = 0.01.

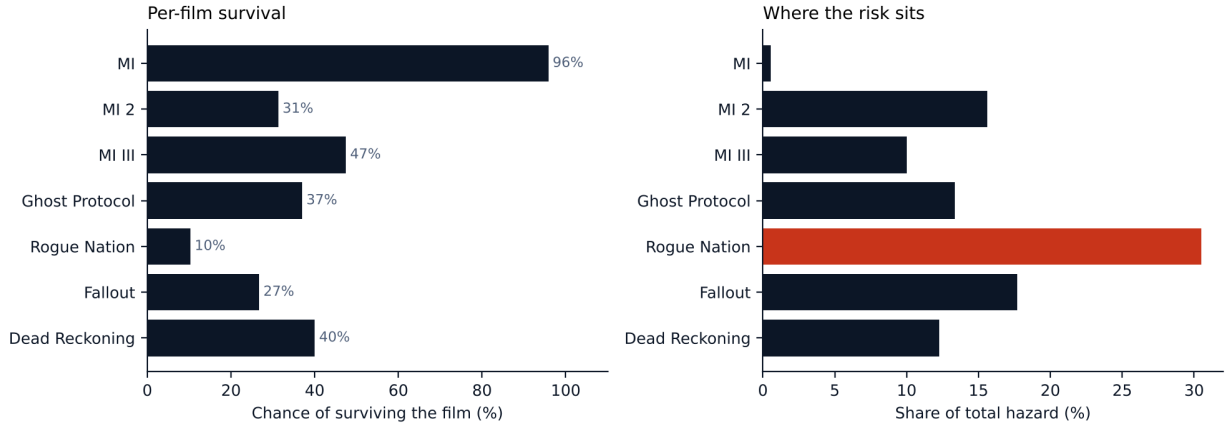


Figure 2. Per-film survival and each film's share of total hazard. Rogue Nation is the most dangerous because of the breath-hold and the chase collisions.

Vehicle collisions account for 77% of the total hazard, followed by the breath-hold (8%) and explosions (6%). Falls, wall slams, fights and fixed-rate events together add about 9%. The model is not saying fights and falls are harmless. It says that vehicle impacts at the speeds I assumed are far deadlier per event than anything else on the list.

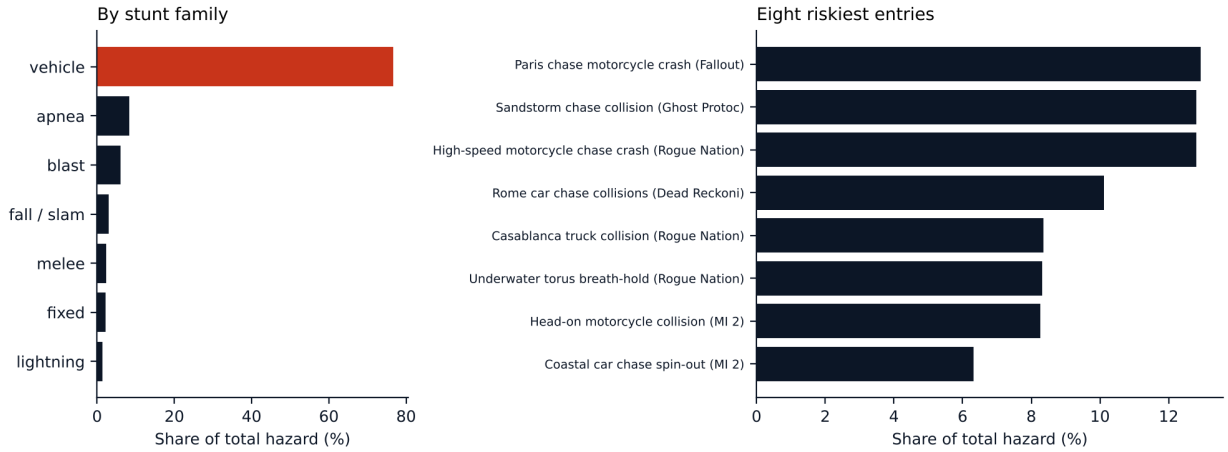


Figure 3. Share of total hazard by stunt family (left) and the eight riskiest individual entries (right).

Entry	Film	Per event	Count	Share
Paris chase motorcycle crash	Fallout	60.8%	1	12.9%
Sandstorm chase collision	Ghost Protocol	60.8%	1	12.8%
High-speed motorcycle chase crash	Rogue Nation	60.8%	1	12.8%
Rome car chase collisions	Dead Reckoning	30.7%	2	10.1%
Casablanca truck collision	Rogue Nation	45.3%	1	8.4%

Table 2. Five riskiest entries.

6. Uncertainty

The median trial (1 in 64,393) is far worse than the point estimate (1 in 1,716). The curves are steep, so a symmetric error in speed or distance pushes some events toward certain death while pulling others only slightly toward safety, and the sum of hazards moves up. The spread is wide: the central half of trials runs from 1 in 9,234 to 1 in 721,123, and only 4.7% of trials give odds better than 1 in 1,000. Halving the noise gives a median of 1 in 3,694; widening it by half gives 1 in 7,727,211.

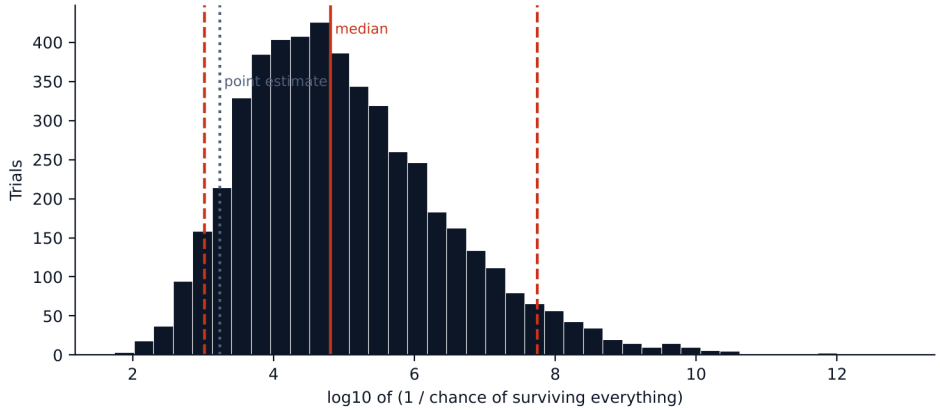


Figure 4. Distribution of combined odds against survival across 5,000 perturbed trials. Solid red is the median, dashed red the 5th and 95th percentiles, dotted grey the point estimate.

7. Sensitivity

Each input group was changed on its own while the others stayed at baseline. The result is most sensitive to vehicle impact speeds: raising them by a quarter worsens the odds by about 3.0 orders of magnitude, and lowering them improves the odds by 1.6. The fatality midpoint of the vehicle curve and the overall event counts come next. Blast sizes, fall parameters, accumulated damage, fixed-rate probabilities and the fight and breath-hold assumptions each move the answer by less than one order of magnitude. Any serious refinement should start with the collision speeds.

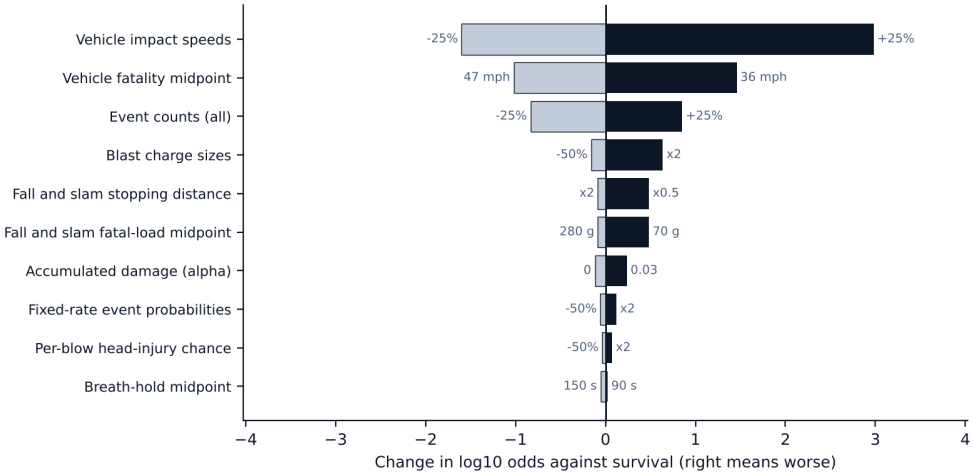


Figure 5. One-at-a-time sensitivity of log10 odds to each input group. Right means worse odds.

8. Limitations

- Event counts, collision speeds, charge sizes and distances come from memory of the films. None was checked against footage, and all rows are marked unverified.
- The curves are stylized and not fitted to data. Vehicle-strike data come from pedestrians, while Hunt is usually in a vehicle or on a motorcycle, where protection and impact geometry differ.
- Fatality is treated as the only outcome. Injury is ignored, so the model cannot say how much damage he would have carried between films.
- Films are independent blocks. There is no healing, treatment or recovery time, and the accumulation rule is a single linear factor.
- Many stunts are not modeled at all, such as gunfire, helicopter rotor strikes, electrocution and high-altitude exposure.
- Event counts are treated as exact in the point estimate, and uncertainty enters only in the trial run.

9. Data and code availability

The dataset (data/ethan_stunts.csv), the interactive simulator (index.html), the JavaScript and Python engines (src/engine.js, src/simulate.py), the analysis and figure scripts (src/analysis.py) and the numerical results (results/results.json) are in the repository. tests/crosscheck.js confirms that the two engines agree on the same seed. The code is released under the MIT license.

References

None of these has been checked. They are recalled from memory and need verification before publication.

[1] Kinney GF, Graham KJ. Explosive Shocks in Air. 2nd ed. Springer; 1985. (Overpressure scaling formula.)

[2] Tefft BC. Impact speed and a pedestrian's risk of severe injury or death. Accident Analysis and Prevention. 2013;50:871-878. (Vehicle-strike curve shape.)

[3] Westman A, Rosen M, Berggren P, Bjornstig U. Parachuting from fixed objects: descriptive study of 106 fatal events in BASE jumping 1981-2006. British Journal of Sports Medicine. 2008. (BASE jump fatality rate.)

[4] Cooper MA, Holle RL. Reducing Lightning Injuries Worldwide. Springer; 2019. (Share of lightning-strike victims who die, about 10%.)

[5] Fall fatality midpoint of roughly four storeys: widely quoted, and no primary source has been checked.